Hangzhou Hikrobot Technology Co., Ltd.

Hikrobot Device Diagnostic Tool

User Manual

HIKROBOT

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Chapter 1 Legal Information

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About this Manual

The Manual includes instructions for using and managing the Product. Pictures, charts, images and all other information hereinafter are for description and explanation only. The information contained in the Manual is subject to change, without notice, due to firmware updates or other reasons. Please find the latest version of this Manual at the Hikrobot website (***https://en.hikrobotics.com***).
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Chapter 2 Disclaimer

The manual guides you to establish and configure the Software. To ensure the properness of usage and stability of the Software, refer to the contents below and read the manual carefully before installation and operation.
As GIFs and videos cannot be played in PDF document, you are recommended to read the online document.

2.1 Get Support

You can get support from us in the following ways:
- **Official Website**: visit ***https://en.hikrobotics.com/*** to get other related documents.
- **Email**: global.support@hikrobotics.com.

2.2 Symbol Conventions

The symbols that might be found in this document are defined as follows.

Symbol	Description
Note	Provides additional information to emphasize or supplement important points of the main text.
Caution	Indicates a potentially hazardous situation which, if not avoided, could result in equipment damage, data loss, performance degradation, or unexpected results.
Danger	Indicates a hazardous situation which, if not avoided, will or could result in death or serious injury.

Chapter 3 Release Notes

Key updates of each version are as follows.

V4.4.0

This version was released on August, 2024.
Adds **Stop when occur** and **Timer period type** in exception IO locating. See ***IO WaveForm Detection***.

V4.2.0

This version was released on March, 2023
This is the first version of Device Diagnostic Tool. The following are the main features.

- Supports diagnostic event overview. See ***Event Overview***.
- Supports detailed event diagnosis. See ***Event Diagnostic***.
- Supports viewing IO waveforms. See ***IO WaveForm Detection***.
- Supports log management. See ***Log Management***.

Chapter 4 Tool Introduction

Device Diagnostic Tool (hereinafter simplified as "the Tool") can monitor the running status of devices in real time, and display the event overview, event chart, IO waveform, and log records.

Supported Device

Frame Grabber Type	Frame Grabber Type Sorted by Interface Standard
Image Frame Grabber	Camera Link frame grabber, Coax Press frame grabber, GigE frame grabber, 10 GigE frame grabber, USB 3.0 frame grabber, and fiber port frame grabber.
Embedded Frame Grabber	GigE frame grabber and USB 3.0 frame grabber.

Main Interface

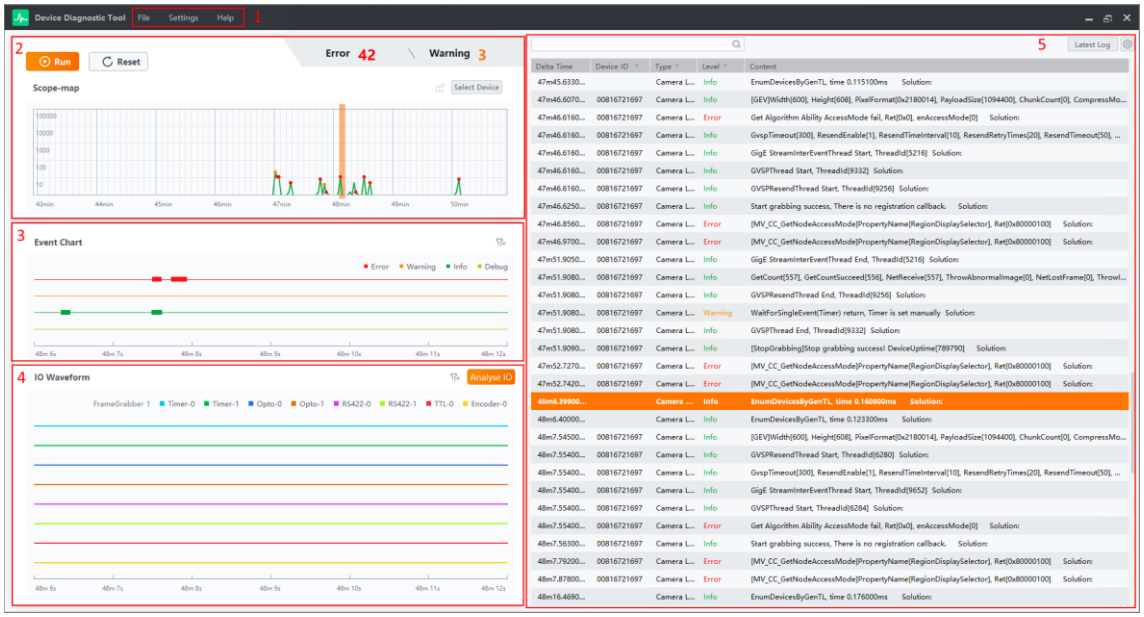

Figure 4-1 Main Interface

No.	Name	Description
1	Menu Bar	View the following functionalities. ● Support importing and exporting log files. ○ Click **File** → **Import** to import a log file from the local PC. ○ Click **File** → **Export** to export a log file to the local PC.

No.	Name	Description
		Note If **Auto Save File** is enabled, exporting log files is not supported. For details, see ***Log Management***. ● Support general settings of log saving. For details, see ***Log Management***. ● Support switching language of the Tool, and viewing the user manual and version information.
2	Event Overview	View the scope-map of frame grabber / camera events. For details, see ***Event Overview***.
3	Event Chart	View the event chart of frame grabber / camera events diagnosed in the selected 6 seconds. For details, see ***Event Diagnostic***.
4	IO Waveform	View the real-time status of IO pulses of frame grabbers / cameras and generate IO analysis. For details, see ***IO WaveForm Detection***.
5	Log Management	View log and event information of frame grabbers / cameras in real time. For details, see ***Log Management***.

Chapter 5 Event Overview

You can take an overview of real-time frame grabber / camera events.

Note

Make sure you have connected frame grabber(s) / camera(s) in advance.

Click **Run**, and a scoped-map will be generated.

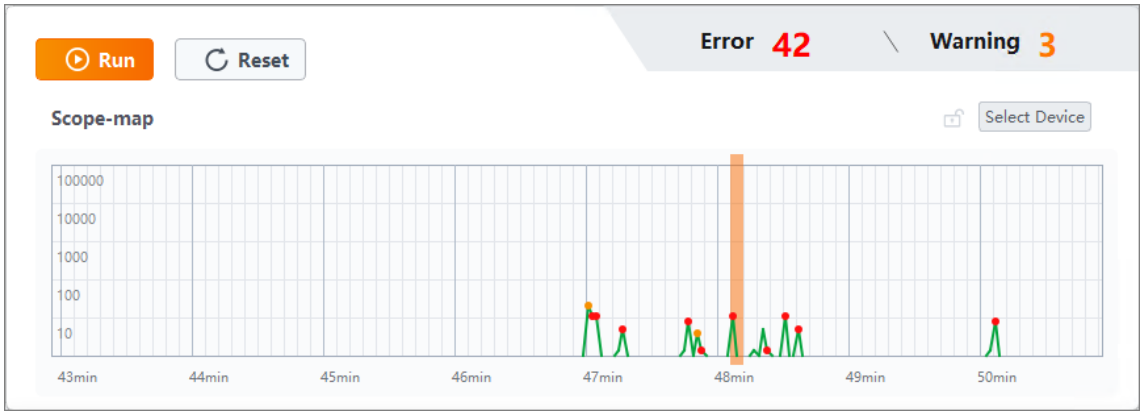

Figure 5-1 Event Overview

You can right-click and drag the cursor on the scope-map to view the events before/after the current time period.

Click to select a time period in the scope-map to view the diagnostic result of events in 6 seconds. For details, see ***Event Diagnostic***.

Area	Description
X-Axis	Timeline. Unit: min.
Y-Axis	Number of events.
Green Point	Normal events.
Red Point	Error events, such as buffer exception, camera offline, and DDR initialization failed. For details, see *Frame Grabber SDK Developer Guide*. **Note** The number of error events is displayed in the upper-right corner.
Orange Point	Warning events such as no parameter collected by frame

Area	Description
	grabber(s) / camera(s). **Note** The number of warning events is displayed in the upper-right corner.
Pause	Click **Pause** the current running. **Note** The Tool will stop receiving data collected from the device.
Reset	Click **Reset** to clear the current data and start running.
Select Device	Click **Select Device** to select frame grabber(s) / camera(s) and click **OK**. **Note** If you select multiple devices, the Tool will only generate a broken line containing all device events diagnosed.
Lock/Unlock Scope Map	Click to lock the current scope-map and stop displaying the new timeline and event data. Click to unlock the current scope-map and display the event data received during locking and newly parsed event data.

Chapter 6 Event Diagnostic

You can view the recorded events of the selected time period in an event chart and view event levels of 6-second events.
If you select a time point in the scope-map, you can view the corresponding event chart of the diagnostic events recorded in 6 seconds.

Figure 6-1 Event Chart

Operation	Description
Zoom In/Out	Scroll the mouse to zoom in/out the X axis (timeline) of the event chart.
Adjust Timeline	Right-click and drag the cursor on the event chart to view the events before/after the current timeline.
View Details	Move the cursor on an event point (rectangle in red, orange, green, or yellow) to view the log information. For details, see ***Log Management***.
Filter by Level	Click to filter the events by Error, Warning, Info, and Debug.

Click to select an event point in the event chart, the corresponding event in IO waveform will also be selected. For details, see***IO WaveForm Detection***.

Chapter 7 IO WaveForm Detection

After an event in the event chart is parsed, you can view the corresponding IO waveform in the IO waveform chart and the number of pulses can be calculated in the real time.

Select IO

For events selected and parsed in the scope map and event chart, you can click in the IO waveform chart to select IO waveform channels and click **OK**.

Note

The IO waveform chart supports displaying maximum 8 channels concurrently.

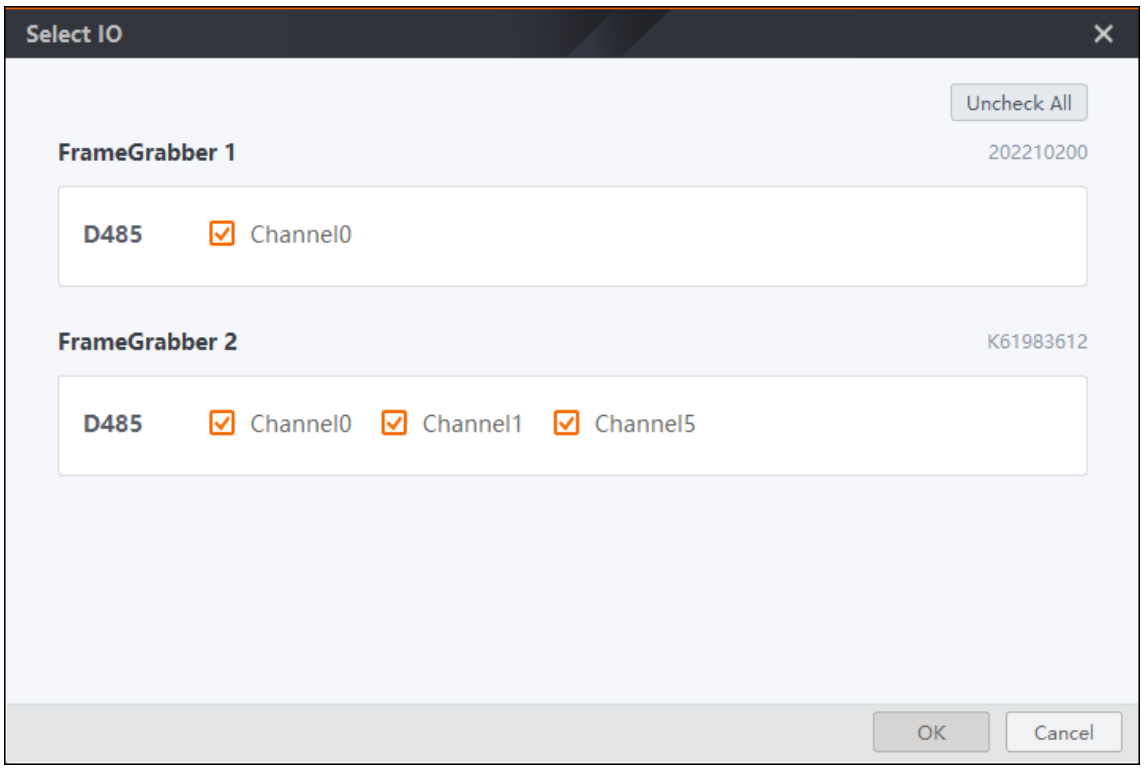

Figure 7-1 Select IO

The IO waveform chart provides a clear view of each IO triggering status. Click **Analyse IO** → **View Pulse Number** and hover your mouse cursor over the IO waveform to view the number of pulses and log/event information of the corresponding IO waveform. You can click **Exit** if pulse number statistics is not needed.

View IO Waveform

Click at any time point in the IO waveform chart, and you can drag the mouse wheel up to

zoom in the chart and down to restore. You can also hold the right mouse button and drag the mouse to view the IO waveform at an earlier/later time.

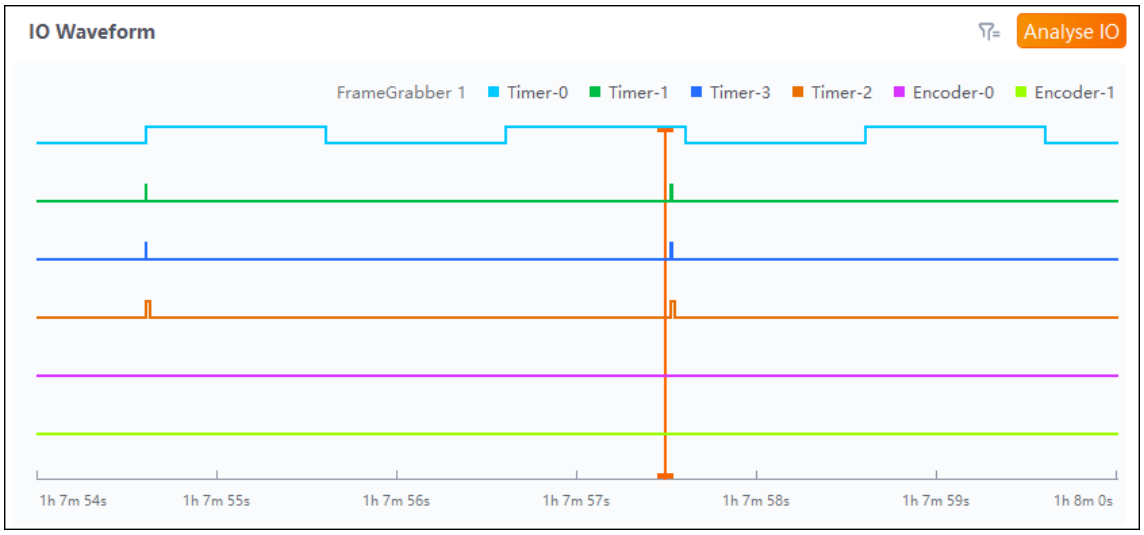

Figure 7-2 IO Waveform Monitoring

Locate Exception IO

Click **Analyse IO** → **Locate Exception IO** to locate the exception IO in the IO waveform chart.

On the **Locate Exception IO** pane, you can specify **Select IO**, **Timer period type** and **Normal Timer period(us)** to locate satisfied exceptions.

Note

After you enable **Stop when occur**, when an exception occurs, the Tool notifies an IO exception occurs with which channel. Then, the Tool pauses logging device status.

To locate one exception IO in the event chart and IO waveform, click **Locate** of the corresponding exception in the Operation column.

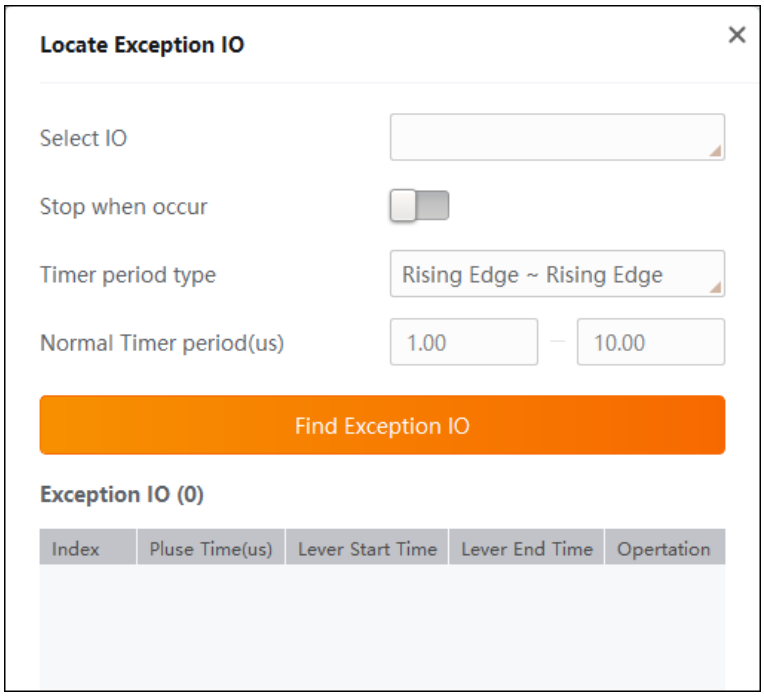

Figure 7-3 Locate Exception IO

Chapter 8 Log Management

After detecting frame grabber / camera events, the Tool can diagnose and accurately locate the abnormal logs in the log list. The tool can be used to monitor and analyze the abnormal events.

General Log Settings

You need to first set log retention period, whether to enable auto save, and auto saving path.

Click **Settings** → **General** in the menu bar.

Log Retention Period

Set the maximum log retention period as the upper limit of log time, which is in minutes. Logs exceeding the set time will not be displayed in the interface.

Note

The parameter should be no more than 100 minutes.

Auto Save File

Enable the parameter to automatically save log files to the set path.

Note

Exporting log file is not allowed when **Auto Save File** is enabled.

Saving Path

You can customize the path of log files. Click to set the path. If no path is selected, files will be saved to the default path: (..\MVS\DeviceDiagnostic\AutoSaveLog).

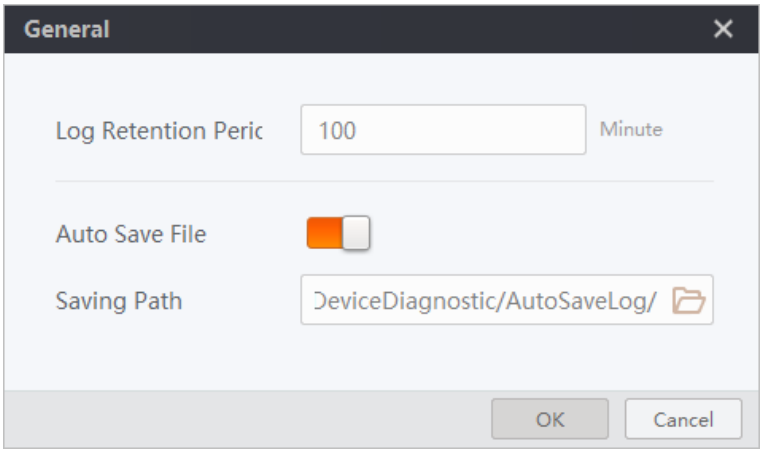

Figure 8-1 General Settings

Log Monitoring

You can monitor real-time log events of frame grabbers / cameras, filter logs, configure log titles, and view all log events within the log retention period.
To quickly locate the events on the scope map, you can click an abnormal event in the scope map, and the corresponding event in the log list can be highlighted (only when the selected "6 Second" event has one exception or warning).
To quickly locate the events on the event chart, you can click any event in the event map to highlight the corresponding event in the log list.

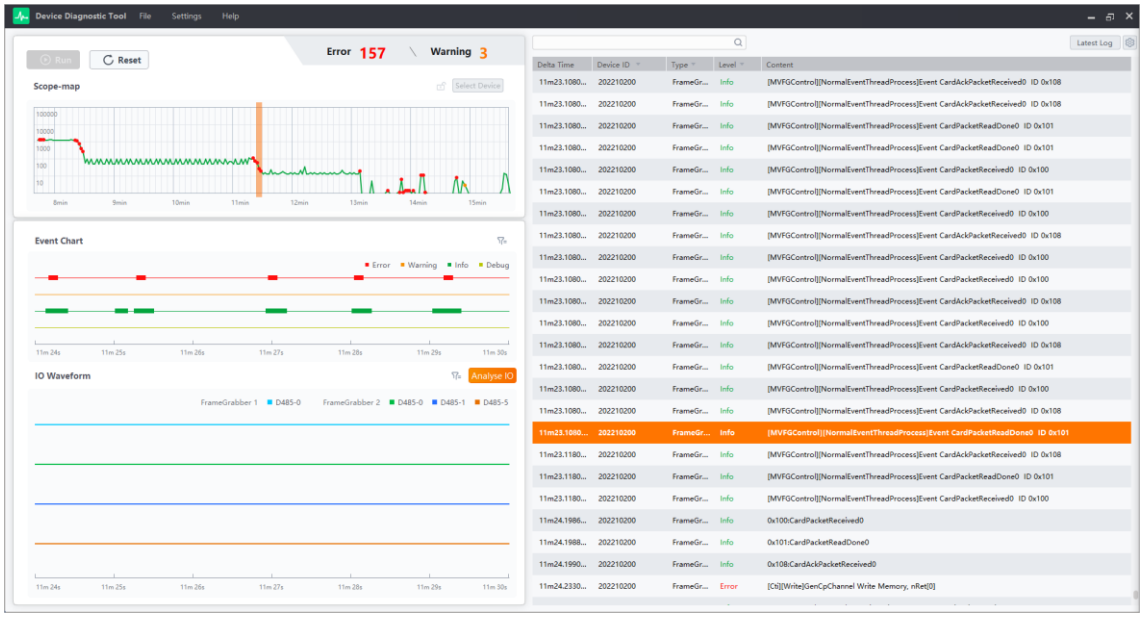

Figure 8-2 View Logs

You can set the columns that will be displayed in the log list.

- Customize the contents of log list: Click ⚙ to customize the columns that will be displayed in the log list. You can check the item(s) of interest to set the contents.

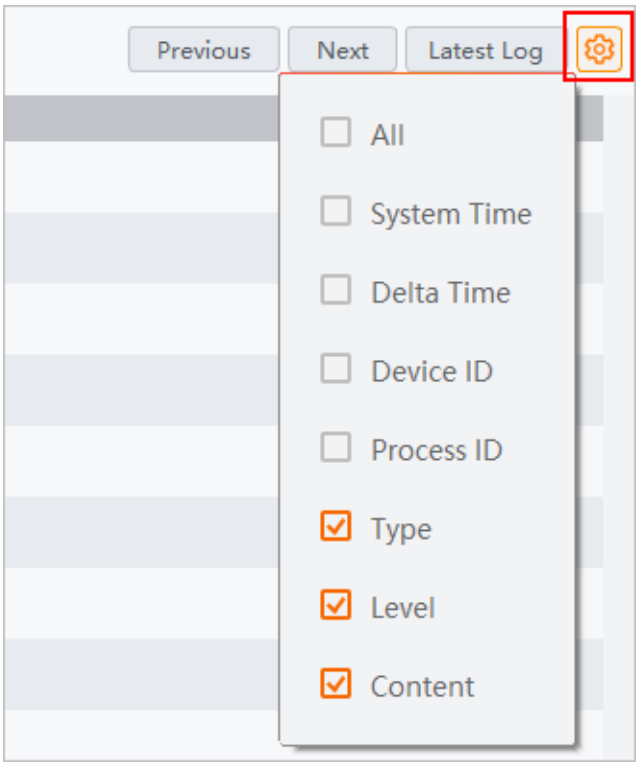

Figure 8-3 Customize Contents of Log List

- Filter log information: You can enter keywords, and click to filter logs. You can also click **Previous**/**Next** to view log information.
- View latest log: Click **Latest Log** to view the latest logs.

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